

# Polynomial Regression

[https://github.com/Kulkarnikeyur/ML\\_assignment.git](https://github.com/Kulkarnikeyur/ML_assignment.git)  
BT2024025

## 1. Introduction

The assignment is to obtain accurate predictions on the provided test datasets using only polynomial regression. Phase 1 predicts the Net Power Score of a multi-stage steam turbine from 6 parameters, while Phase 2 predicts the Thermal Anomaly Score at locations in a 3-dimensional subterranean geothermal reservoir.

The complete modelling flow is

Train Data  $\rightarrow$  PolynomialFeatures  $\rightarrow$  StandardScaler  $\rightarrow$  L1/L2 Regularization  
 $\rightarrow$  5-Fold Cross-Validation  $\rightarrow$  Compare CV RMSE  $\rightarrow$  Select Best (Regularization, Degree,  $\alpha$ )  
 $\rightarrow$  Retrain on Full Train Data  $\rightarrow$  Test Data  $\rightarrow$  Predictions

The final model is selected using the lowest cross-validation RMSE.

## 2. Methodology

### 2.1 Polynomial Regression

Polynomial regression extends linear regression by transforming the input variables into polynomial features. For example, for two variables and degree two, the transformed features include

$$x_1, \quad x_2, \quad x_1^2, \quad x_1x_2, \quad x_2^2.$$

The general polynomial model is

$$\hat{y} = \beta_0 + \sum_i \beta_i x_i + \sum_{i,j} \beta_{ij} x_i x_j + \dots$$

where all terms have total degree up to the selected degree.

### 2.2 Regularization

Two regularization methods are evaluated.

L1 regularization (Lasso), which can drive some coefficients to zero

$$\min_{\beta} \left[ \frac{1}{2n} \sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j |\beta_j| \right].$$

L2 regularization (Ridge), shrinks coefficients and is useful when polynomial features are correlated

$$\min_{\beta} \left[ \frac{1}{2n} \sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j \beta_j^2 \right].$$

### 2.3 Cross-Validation and Model Selection

Phase 1 evaluates polynomial degrees 1–10, while Phase 2 evaluates degrees 1–20. For every degree, both L1 and L2 regularization are tested with

$$\alpha \in \{0.0001, 0.001, 0.01, 0.1, 1, 10, 100\}.$$

Thus, 140 configurations are evaluated for Phase 1 and 280 configurations for Phase 2.

5-fold cross-validation is performed for every configuration. The scaler is fitted only on the training portion of each fold to avoid data leakage.

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \text{RMSE} = \sqrt{\text{MSE}},$$

and

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

The configuration with the minimum CV RMSE is selected and subsequently retrained on the complete training dataset.

### 3. Phase 1: Power Plant Steam Turbine Optimization

#### 3.1 Problem Setup

Phase 1 predicts the Net Power Score of a multi-stage steam turbine from six operational parameters:

- $x_1$ : High-pressure steam valve adjustment,
- $x_2$ : Condenser coolant flow rate adjustment,
- $x_3$ : Re-injection pump hydraulic pressure,
- $x_4$ : Turbine blade pitch angle,
- $x_5$ : Non-condensable gas exhaust valve rate,
- $x_6$ : Steam inlet pressure adjustment.

The regression problem is :

$$y = f(x_1, x_2, x_3, x_4, x_5, x_6).$$

#### 3.2 Model Selection

The best configuration obtained after evaluating all degrees, regularization types and alpha values was

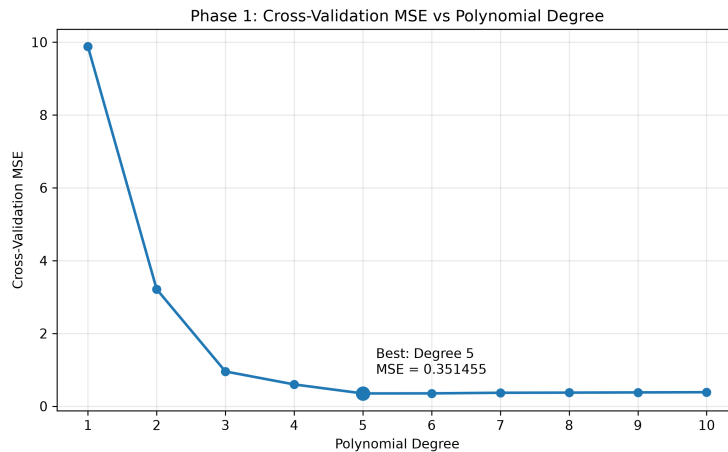
|                    |             |                 |
|--------------------|-------------|-----------------|
| L1 regularization, | Degree = 5, | $\alpha = 0.01$ |
|--------------------|-------------|-----------------|

with the following five-fold cross-validation performance:

| Metric   | Value    |
|----------|----------|
| CV MSE   | 0.351455 |
| CV RMSE  | 0.592837 |
| CV $R^2$ | 0.967823 |

Table 1: Best Phase 1 cross-validation performance.

#### 3.3 MSE vs Degree plot analysis



The CV MSE decreases rapidly as the degree increases from 1 to 5. The minimum occurs at degree 5. Increasing the degree beyond 5 does not improve validation performance. As you can see in the plot above the MSE remains slightly higher for subsequent degrees.

This shows that polynomial complexity must be controlled rather than simply increased. The regularization search is also important because the best model is determined jointly by degree, regularization type and  $\alpha$ .

### 3.4 Final Model

After selecting the configuration, the L1-regularized model  $\alpha = 0.01$  was retrained on the complete Phase 1 training dataset using standardized, degree-5 polynomial features.

The trained model was then applied to the Phase 1 test dataset to generate the final predictions.

The selected fifth-degree model provides a strong validation fit, with  $R^2 = 0.967823$ . The significantly lower error highlights strong nonlinearity between the six operational parameters and the Net Power Score.

## 4. Phase 2: Subterranean Thermal Reservoir Mapping

### 4.1 Problem Setup

Phase 2 predicts the Thermal Anomaly Score at locations in a three-dimensional geological region. The three input variables are

- $x_1$ : East-West coordinate offset,
- $x_2$ : North-South coordinate offset,
- $x_3$ : Vertical depth offset relative to the basecamp.

The regression problem is

$$y = f(x_1, x_2, x_3).$$

### 4.2 Model Selection

After evaluating all 280 combinations of degree, regularization type and alpha, the best configuration was

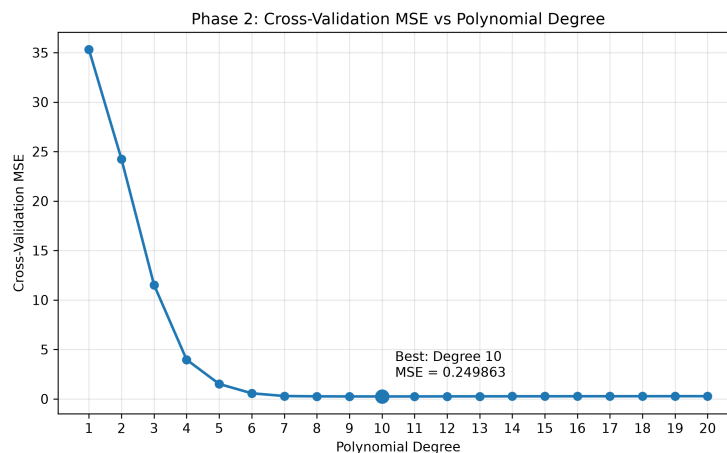
|                    |              |              |
|--------------------|--------------|--------------|
| L2 regularization, | Degree = 10, | $\alpha = 1$ |
|--------------------|--------------|--------------|

with

| Metric   | Value    |
|----------|----------|
| CV MSE   | 0.249863 |
| CV RMSE  | 0.499863 |
| CV $R^2$ | 0.994487 |

Table 2: Best Phase 2 cross-validation performance.

### 4.3 MSE vs Degree plot analysis



The validation error decreases strongly from degree 1-10. The minimum CV MSE is obtained at degree 10. Beyond this point, the MSE starts increasing, showing that additional polynomial complexity does not improve generalization.

## 4.4 Final Model

After configuration selection, a Ridge model  $\alpha = 1$  was retrained on the complete Phase 2 training dataset using standardized, degree-10 polynomial features.

The final model was then applied to the Phase 2 test coordinates.

This achieves a CV  $R^2$  of 0.994487, indicating that the selected model captures the nonlinear relationship between the three spatial coordinates and the Thermal Anomaly Score very effectively.

## 5. Comparison and Conclusion

### 5.1 Final Model Comparison

| Phase   | Reg. | Degree | $\alpha$ | MSE      | RMSE     | $R^2$    |
|---------|------|--------|----------|----------|----------|----------|
| Phase 1 | L1   | 5      | 0.01     | 0.351455 | 0.592837 | 0.967823 |
| Phase 2 | L2   | 10     | 1.00     | 0.249863 | 0.499863 | 0.994487 |

Table 3: Final selected models and five-fold CV performance.

The two phases require different model configurations. Phase 1 achieves its best performance using L1 regularization and degree 5, whereas Phase 2 performs best using L2 regularization and degree 10.

### 5.2 Conclusion

The experiments demonstrate that polynomial degree, regularization type and regularization strength should be selected jointly rather than choosing polynomial degree alone.

The final Phase 1 model is

$$\boxed{\text{L1} + \text{Degree 5} + \alpha = 0.01}$$

with CV RMSE = 0.592837 and CV  $R^2$  = 0.967823.

The final Phase 2 model is

$$\boxed{\text{L2} + \text{Degree 10} + \alpha = 1}$$

with CV RMSE = 0.499863 and CV  $R^2$  = 0.994487.

The MSE-versus-degree plots show that increasing polynomial complexity initially improves validation performance, but after an optimal degree, additional complexity no longer improves generalization.

Finally, the selected models were retrained on the complete training datasets and used to generate predictions for the respective test datasets.